



SCHOOL OF COMPUTER SCIENCE & ENGINEERING

PROJECT APPROVAL FORM AND ABSTRACT

Odd 2026-2027

B. Tech Project Details

Project ID: _____

Title	Reliable Breast Ultrasound Classification Using Uncertainty-Aware Selective Deep Learning		
ProjectType	☐ Community based design problem (Interdisciplinary) ☐ App Development / Utility ☐ IOT/Hardware based ☐ AI/ML/Data Science ☐ Healthcare Projects	ProjectOutcome	☐ Project and Research Paper ☐ Project and Patent ☐ Project and Book Chapter
		SDG GoalNumber	3
Publication Target	☐ SCOPUS Journal ☐ SCOPUS Conference ☐ SCOPUS Book Chapter ☐ SCI Journal	Guide Name: Mr.ARUNENDRA MANI TRIPATHI	

Student Details:

S. No	Name	Enrollment Number	Admission Number	Program / Branch	Sem
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Sustainable Development Goals (SDG Goals), Targets and Indicators

- Goal 1: End poverty in all its forms everywhere
- Goal 2: End hunger, achieve food security and improved nutrition and promote sustainable agriculture
- Goal 3: Ensure healthy lives and promote well-being for all at all ages
- Goal 4: Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all
- Goal 5: Achieve gender equality and empower all women and girls
- Goal 6: Ensure availability and sustainable management of water and sanitation for all
- Goal 7: Ensure access to affordable, reliable, sustainable and modern energy for all
- Goal 8: Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all
- Goal 9: Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation
- Goal 10: Reduce inequality within and among countries
- Goal 11: Make cities and human settlements inclusive, safe, resilient and sustainable
- Goal 12: Ensure sustainable consumption and production patterns
- Goal 13: Take urgent action to combat climate change and its impacts
- Goal 14: Conserve and sustainably use the oceans, seas and marine resources for sustainable development
- Goal 15: Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss
- Goal 16: Promote peaceful and inclusive societies for sustainable development, provide access to justice for all and build effective, accountable and inclusive institutions at all levels
- Goal 17: Strengthen the means of implementation and revitalize the Global Partnership for Sustainable Development

[Reliable Breast Ultrasound Classification Using Uncertainty-Aware Selective Deep Learning]

Area/Domain of Project: AIML / Healthcare

ABSTRACT:

Breast cancer remains one of the most prevalent and life-threatening diseases affecting women worldwide, making early and accurate diagnosis critical for effective treatment. Ultrasound imaging is a widely used, non-invasive modality for breast tumor assessment, and deep learning models have shown promising results in automatically classifying ultrasound images as benign or malignant. However, while existing deep-learning-based classification systems can achieve high accuracy, accuracy alone does not indicate whether an individual prediction can be trusted — a significant limitation in a clinical setting where incorrect confident predictions can have serious consequences. This project, titled “Reliable Breast Ultrasound Classification Using Uncertainty-Aware Selective Deep Learning,” builds upon an existing IEEE-based deep learning classification approach and extends it by incorporating uncertainty-aware selective prediction. The proposed system estimates the confidence associated with each prediction and classifies breast ultrasound images automatically only when the model is sufficiently confident; cases with high uncertainty are instead flagged and deferred to expert radiologists for manual review. By combining classification performance with a reject option based on predictive uncertainty, the system aims to improve not only diagnostic accuracy but also the overall reliability, safety, and clinical usability of AI-assisted breast ultrasound analysis. The project emphasizes that trustworthy medical AI systems must know when they are unsure, rather than producing overconfident predictions on ambiguous cases.

Problem Statement:

Deep learning models for breast ultrasound image classification can achieve high overall accuracy, but they typically provide a prediction for every input without indicating how confident that prediction actually is. In clinical practice, this is problematic because a model may generate an incorrect classification with high confidence, potentially leading to misdiagnosis if the prediction is trusted blindly. Existing approaches largely focus on improving classification accuracy while overlooking the reliability of individual predictions, leaving a gap between model performance on benchmark datasets and safe, practical deployment in real diagnostic workflows. There is a need for a system that not only classifies breast ultrasound images accurately but also recognizes when it is uncertain, so that ambiguous or difficult cases can be flagged for expert review rather than being misclassified with false confidence. This project addresses this gap by developing an uncertainty-aware selective deep learning system for breast ultrasound classification that balances automated prediction with human oversight on uncertain cases.

Signature of Students

Signature of Guide

